Mathematical methods for scientific computing
Partial differential equations

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Partial differential equations

- The overall field of partial differential equations is large, you have likely covered many different techniques for solving these equations, e.g. separation of variables
- We are not going to cover that here
- Instead, we are going to focus on how PDEs are used for scientific computing
- A wide range of physical problems can be expressed as the solutions to PDEs, often very complex systems of PDEs
- In these cases, we would expect our computational methods to do the actual solving
- The challenge might often be expressing our system in a manner that can be solved - we need to know something about the behaviour of our system

Outline

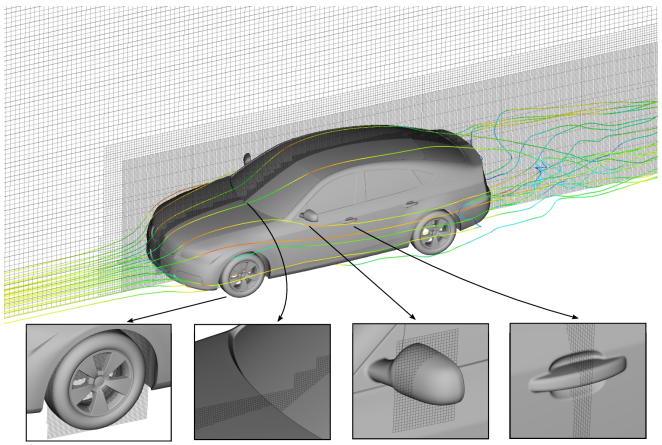
1. Classification of PDEs
2. Different forms for PDEs
3. Discontinuities

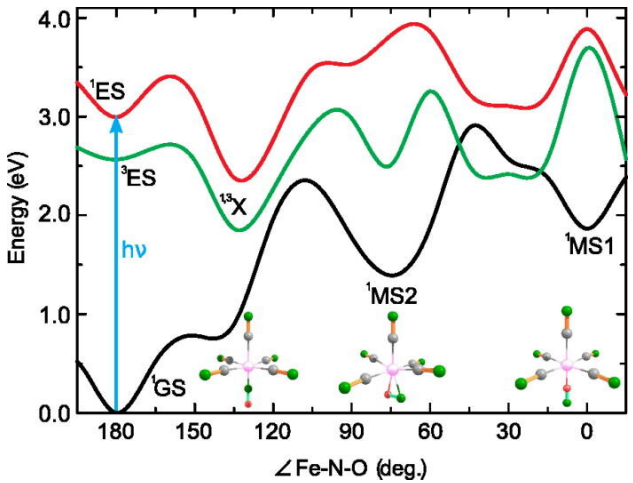
Classification of PDEs

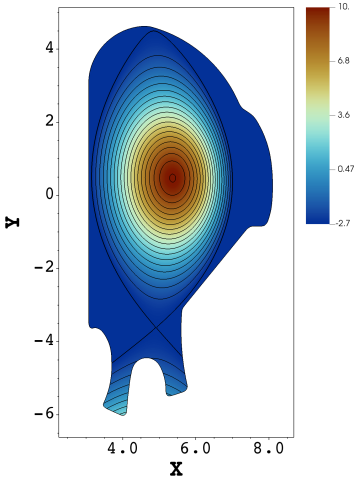
- There is no single technique for solving a PDE, or system of PDEs
- This is based on the mathematical and physical behaviour of the underlying problem
- By classifying our PDEs (a mathematical approach), we can identify whether they are capable of producing the correct physics we wish to capture
- And we can also ensure we can identify a computational approach which will allow this behaviour to be captured
- More accurately, we can identify a computational approach that is both suitably accurate and suitably efficient for the problem

Steady state problems

- One class of problem could be those where the solution is a fixed value (typically in time)
- This could be considered to be a balance of physical states, e.g.
  - Airflow around an object, e.g. a car or aerofoil - from steady state conditions, lift and drag can be computed
  - Minimum energy states for a molecular configuration
  - Plasma equilibrium configurations in a tokamak reactor - perfect balance between pressure gradients and magnetic field

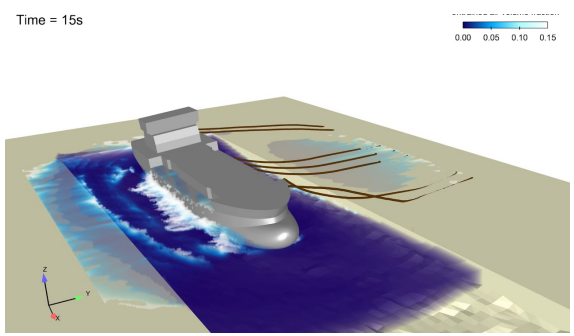


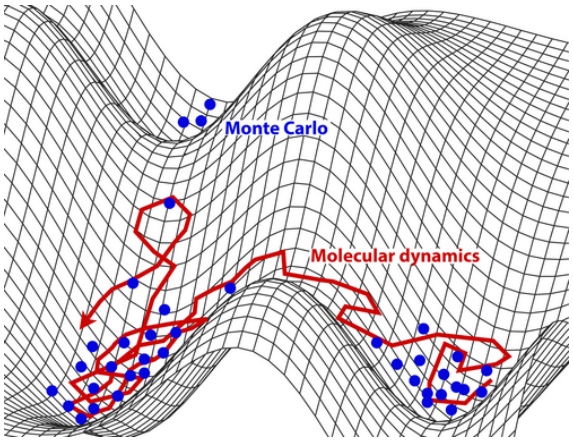


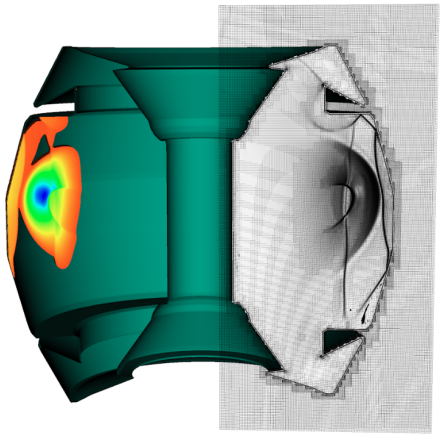


Time-varying problems

- Alternatively, our solution may continuously change in space and time
- Here, examples include
  - Water flow around an object as it moves through the real world
  - Evolution of atoms according to molecular dynamics
  - Plasma instabilities in a reactor - perfect balance is not physically attainable







Mathematical classification

- Whilst steady state and time-varying do describe features of the equations, this doesn't yet help us decide upon our numerical method
- Mathematically, we consider **speeds of information propagation**
  - Equations where **all** information propagates infinitely fast
  - Equations where **some** information propagates infinitely fast, but some travels with a finite speed
  - Equations where **all** information travels at a finite speed

Mathematical classification

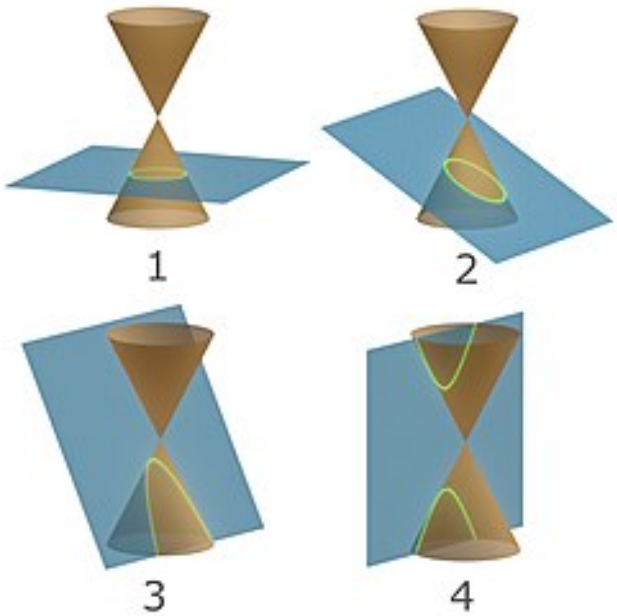
- Whilst steady state and time-varying do describe features of the equations, this doesn't yet help us decide upon our numerical method
- Mathematically, we consider **speeds of information propagation**
  - Equations where **all** information propagates infinitely fast
    Elliptic equations
  - Equations where **some** information propagates infinitely fast, but some travels with a finite speed
    Parabolic equations
  - Equations where **all** information travels at a finite speed
    Hyperbolic equations
- The behaviour in parabolic equations is often also referred to as **diffusion**

Classification - conic sections

- Classification of PDEs is actually a wide area with ongoing research - we shall only consider a few exampes
- One example starts with drawing comparisons to conic sections (the lines through which a cone can be cut)
- Such lines are parameterised through

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

- The sign of the discriminant, $\mathcal{D} = B^2 - 4AC$ determines the shape
  - $D < 0$: Ellipse
  - $D = 0$: Parabola
  - $D > 0$: Hyperbola



Classification - linear second-order PDSs

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

- There is a similarity between conic sections and linear second-order PDEs

$$A\frac{\partial^2 u}{\partial x^2} + B\frac{\partial^2 u}{\partial x \partial y} + C\frac{\partial^2 u}{\partial y^2} + D\frac{\partial u}{\partial x} + E\frac{\partial u}{\partial y} + F = 0$$

- We can still consider the discriminant, $\mathcal{D} = B^2 - 4AC$, now we have:
  - $D < 0$: Elliptic equation
  - $D = 0$: Parabolic equation
  - $D > 0$: Hyperbolic equation
- However, this is a very limited set of equations, what happens if you are modelling something that is not two-dimensional, not linear, not second order and/or not a single variable?

Classification - systems

- Full classification of PDEs involves analysis of **characteristics** (which govern information propagation) - this can get complex very quickly
- However, sometimes, equations of the form

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot \mathbf{f}(\mathbf{u}) = 0$$

- This can be written as

$$\frac{\partial \mathbf{u}}{\partial t} + \frac{\partial \mathbf{f}}{\partial \mathbf{u}} \nabla \cdot \mathbf{u} = 0$$

- The eigenvalues of the Jacobian matrix **might** allow us to classify the system of PDEs:
  - All purely complex: Elliptic system
  - All real and distinct: Hyperbolic system
- Other classifications are not so straightforward, and need more investigation - nonlinear equations could sometimes have conditionally real eigenvalues, i.e. some physical situations could provide imaginary values

Outline

1. Classification of PDEs
2. Different forms for PDEs
3. Discontinuities

Different forms of PDEs

- Throughout the continuum courses, you will be looking at solutions to various partial differential equations
- In general, these can all be written in a form which looks like

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot \mathbf{f}(\mathbf{u}) = \mathbf{s}(\mathbf{u})$$

- They will not always look like this, though
- Especially since the term $\mathbf{s}(\mathbf{u})$ can hide anything that is not a time derivative or divergence of a quantity
- However, the physical systems we consider, whether you start from the Navier-Stokes equations, Maxwell's equations or even Einstein's field equations, can be written in this form (in some cases, with considerable effort)

Commonly encountered forms

- For simplicity, we shall consider 1D, and $\mathbf{s} = \mathbf{0}$

$$\frac{\partial u_i}{\partial t} + \frac{\partial f_i(\mathbf{u})}{\partial x} = 0$$

- Which might look like

$$\frac{\partial u_i}{\partial t} + A_{ij}\frac{\partial u_j}{\partial x} = 0$$

- or

$$\frac{\partial w_i(\mathbf{u})}{\partial t} + B_{ij}\frac{\partial w_j(\mathbf{u})}{\partial x} = 0$$

- or

$$\frac{\partial \mathcal{V}_i}{\partial t} + \Lambda_{ij}\frac{\partial \mathcal{V}_j}{\partial x} = 0$$

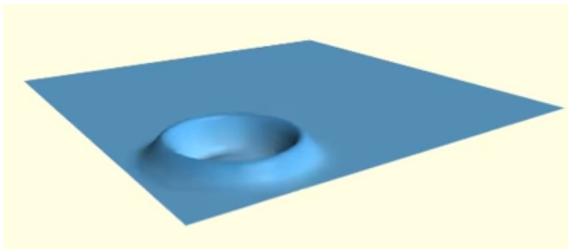
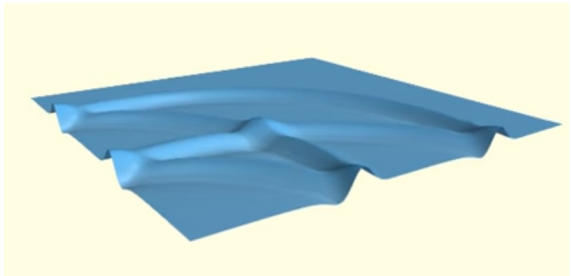
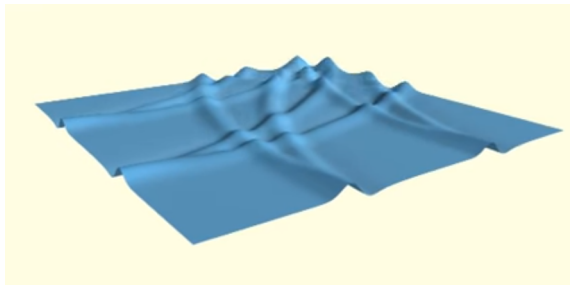
  where Λ is a diagonal matrix

- These forms may allow for accurate numerical solution, analysis of the equations or even identification of analytic solutions

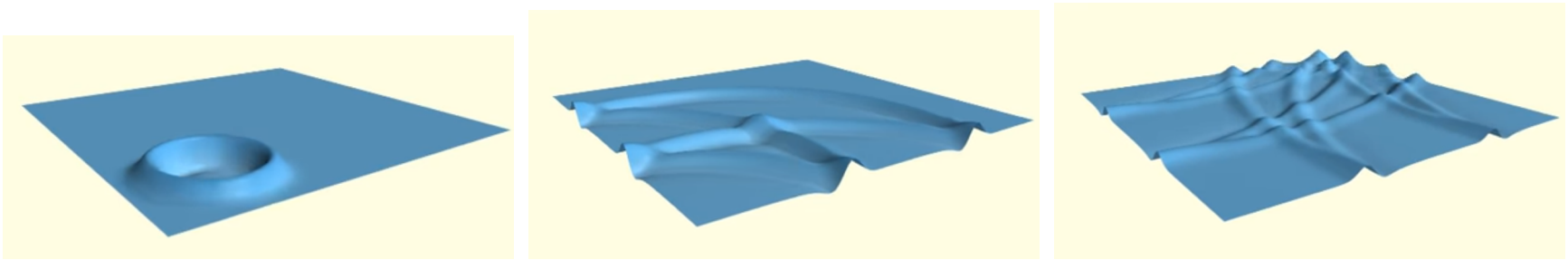
An example - the shallow water equations

- As a simple, but non-trivial example to demonstrate these different forms, and some of the mathematical techniques used, we consider the shallow water equations
- They are a reduction of the Navier-Stokes equations and are a suitable approximation for some systems where there is a vertical length scale much smaller than other length scales (such as shallow water)

$$\frac{\partial}{\partial t}\begin{pmatrix} \phi \\ \phi u \end{pmatrix} + \frac{\partial}{\partial x}\begin{pmatrix} \phi u \\ \phi u^2 + \frac{1}{2}\phi^2 \end{pmatrix} = \begin{pmatrix} 0 \\ g\phi\frac{\partial h}{\partial x} \end{pmatrix}$$

An example - the shallow water equations



$$\frac{\partial}{\partial t}\begin{pmatrix}\phi\\ \phi u\end{pmatrix}+\frac{\partial}{\partial x}\begin{pmatrix}\phi u\\ \phi u^2+\frac{1}{2}\phi^2\end{pmatrix}=\begin{pmatrix}0\\ g\phi\frac{\partial h}{\partial x}\end{pmatrix}$$

- In this one-dimensional form, u is the velocity, h the height (or depth) of the lower surface (e.g. the sea bed) and g is gravitational acceleration
- ϕ is the geopotential, related to the height through $\phi = gH = g(\eta + h)$ where η is the height above an initial surface
- If a uniform depth is considered (which, for laziness, it will be in this lecture), the right hand side vanishes

Conservation form

$$\frac{\partial u_i}{\partial t} + \frac{\partial f_i\left(\mathbf{u}\right)}{\partial x} = 0$$

- This form of equations is referred to as conservation form

$$\frac{\partial}{\partial t}\begin{pmatrix} \phi \\ \phi u \end{pmatrix} + \frac{\partial}{\partial x}\begin{pmatrix} \phi u \\ \phi u^2 + \frac{1}{2}\phi^2 \end{pmatrix} = \mathbf{0}$$

- It is clear that the shallow water equations, as presented so far, are in this form,

$$\mathbf{u} = \begin{pmatrix} \phi \\ \phi u \end{pmatrix} \qquad \mathbf{f} = \begin{pmatrix} \phi u \\ \frac{(\phi u)^2}{\phi} + \frac{1}{2}\phi^2 \end{pmatrix}$$

- Note that ϕ and ϕu are **independent variables** in this case

Conservation form

- The term conservation form is directly related to the fact that in multiple dimensions, our equations look like

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot \mathbf{f}(\mathbf{u}) = \mathbf{0}$$

- One definition of divergence of a vector field is that it measures the rate at which the vector field enters or leaves a volume,

$$\nabla \cdot \mathbf{F} = \lim_{V \to 0} \frac{1}{V} \int_S \mathbf{F} \cdot \hat{\mathbf{n}} \mathrm{d}S$$

- This is **conservation**; the only way to change $\mathbf{u}$ is by adding or removing some from a volume
- Alternatively, the divergence term could be balanced by a term on the right hand side of the equation - we could have **balance law form**

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot \mathbf{f}\left(\mathbf{u}\right) = \mathbf{s}\left(\mathbf{u}\right)$$

Non-conservation form (and the Jacobian)

$$\frac{\partial u_i}{\partial t} + A_{ij}\frac{\partial u_j}{\partial x} = 0$$

- It can be good to have all derivatives being taken of the same vector, ($\mathbf{u}$), this is a non-conservative form
- The matrix quantity A is the **Jacobian** of the vector $\mathbf{f}$

$$A_{ij} = \frac{\partial f_i}{\partial u_j} \qquad \text{and by the chain rule} \qquad \frac{\partial f_i}{\partial x} = \frac{\partial f_i}{\partial u_j}\frac{\partial u_j}{\partial x}$$

- For the shallow water equations, we get

$$\frac{\partial}{\partial t}\begin{pmatrix} \phi \\ \phi u \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ -u^2+\phi & 2u \end{pmatrix}\frac{\partial}{\partial x}\begin{pmatrix} \phi \\ \phi u \end{pmatrix} = \mathbf{0}$$

- Because ϕ and ϕu are independent variables,

$$\frac{\partial(\phi u)}{\partial \phi} = 0$$

Primitive-variable form

$$\frac{\partial w_i\left(\mathbf{u}\right)}{\partial t} + B_{ij}\frac{\partial w_j\left(\mathbf{u}\right)}{\partial x} = 0$$

- Sometimes, the conserved variables are not the most intuitive variables to use, and the shallow water equations are a good example of this
- Why use geopotential if we could use height?
- Alternative variable choices are **primitive variables**, and may allow for intuitive or analytic understanding of the equations
- To qualify as a set of primitive variables, we need variables which allow for a **closed system of equations**, and at least one variable should not also be a conserved variable
- In the shallow water equations, a suitable choice of primitive variables might be total height, H, and velocity, u
- But we could also have ϕ and u, η and u^2 but **not** ϕ and H (from which u cannot be computed)

Primitive-variable form

$$\frac{\partial w_i\left(\mathbf{u}\right)}{\partial t} + B_{ij}\frac{\partial w_j\left(\mathbf{u}\right)}{\partial x} = 0$$

- Obtaining primitive-variable form is now straightforward

$$\frac{\partial u_i}{\partial w_j}\frac{\partial w_j}{\partial t} + \frac{\partial f_j\left(\mathbf{w}\right)}{\partial w_j}\frac{\partial w_j}{\partial x} = 0$$

- Therefore

$$B_{ij} = \left(\frac{\partial u_i}{\partial w_k}\right)^{-1}\frac{\partial f_k\left(\mathbf{w}\right)}{\partial w_j} = \left(\frac{\partial u_i}{\partial w_k}\right)^{-1}\frac{\partial f_k}{\partial u_l}\frac{\partial u_l}{\partial w_j}$$

- This last expression implies A and B are **similar matrices**
- For the shallow water equations,

$$\frac{\partial}{\partial t}\begin{pmatrix} H \\ u \end{pmatrix} + \begin{pmatrix} u & H \\ g & u \end{pmatrix}\frac{\partial}{\partial x}\begin{pmatrix} H \\ u \end{pmatrix} = \mathbf{0}$$

- Often, it is not worth going through the matrix calculations, and it is easy to do this by simple manipulation of the equations

Characteristic form

$$\frac{\partial \mathcal{V}_i}{\partial t} + \Lambda_{ij} \frac{\partial \mathcal{V}_j}{\partial x} = 0$$

- Characteristic form allows for a wide range of analysis of partial differential equations
- We shall see it being used for finding exact solutions in some cases, approximate solutions in others, or understanding the behaviour of the system
- It is important to note that not all equations can be written in characteristic form; since Λ is a diagonal matrix, at some point we will need to diagonalise a matrix
- We could consider characteristic form to be a very particular case of primitive variable form

$$\frac{\partial w_i(\mathbf{u})}{\partial t} + B_{ij} \frac{\partial w_j(\mathbf{u})}{\partial x} = \frac{\partial w_i(\mathbf{u})}{\partial t} + (C_{ik})^{-1} A_{kl} C_{lj} \frac{\partial w_j(\mathbf{u})}{\partial x} = 0$$

Characteristic form

$$\frac{\partial \mathcal{V}_i}{\partial t} + \Lambda_{ij} \frac{\partial \mathcal{V}_j}{\partial x} = 0$$

- We need to find a matrix, C, such that $(C_{ik})^{-1} A_{kl} C_{lj}$ is diagonal

$$\frac{\partial w_i(\mathbf{u})}{\partial t} + (C_{ik})^{-1} A_{kl} C_{lj} \frac{\partial w_j(\mathbf{u})}{\partial x} = 0$$

- In doing so, we will effectively choose the 'primitive' variables for this case, since

$$C_{ij} = \frac{\partial u_i}{\partial w_j} = \frac{\partial u_i}{\partial \mathcal{V}_j}$$

- This may well be familiar, we are **diagonalising the Jacobian** to achieve this
- Therefore the entries of Λ are the eigenvalues of A, and C_{ij} and $(C_{ij})^{-1}$ are matrices of the right and eigenvectors and its inverse respectively

Shallow water equations' eigenvalues and eigenvectors

- Two eigenvalues,

$$\lambda_1 = u + \sqrt{\phi} = u + \sqrt{gH} \qquad \lambda_2 = u - \sqrt{\phi} = u - \sqrt{gH}$$

- These eigenvalues are both **real and distinct** - we know we can write the equation in characteristic form as a result
- The right eigenvectors of B are:

$$\mathbf{r}_1 = \begin{pmatrix} \frac{H}{\sqrt{gH}} \\ 1 \end{pmatrix} \qquad \mathbf{r}_2 = \begin{pmatrix} -\frac{H}{\sqrt{gH}} \\ 1 \end{pmatrix}$$

- The matrix of right eigenvectors is then

$$\frac{\partial \mathbf{w}}{\partial \boldsymbol{\mathcal{V}}} = \begin{pmatrix} \frac{H}{\sqrt{gH}} & -\frac{H}{\sqrt{gH}} \\ 1 & 1 \end{pmatrix}$$

Shallow water equations' eigenvalues and eigenvectors

$$\frac{\partial \mathbf{w}}{\partial \boldsymbol{\mathcal{V}}} = \begin{pmatrix} \frac{1}{g}\sqrt{gH} & -\frac{1}{g}\sqrt{gH} \\ 1 & 1 \end{pmatrix}$$

- The inverse of this matrix is now

$$\left(\frac{\partial \mathbf{w}}{\partial \boldsymbol{\mathcal{V}}}\right)^{-1} = \frac{1}{2}\begin{pmatrix} \frac{\sqrt{gH}}{H} & 1 \\ -\frac{\sqrt{gH}}{H} & 1 \end{pmatrix} \tag{2.1}$$

- We now have, as expected,

$$\Lambda = \begin{pmatrix} u+\sqrt{gH} & 0 \\ 0 & u-\sqrt{gH} \end{pmatrix} = \frac{1}{2}\begin{pmatrix} \frac{\sqrt{gH}}{H} & 1 \\ -\frac{\sqrt{gH}}{H} & 1 \end{pmatrix}\begin{pmatrix} u & H \\ g & u \end{pmatrix}\begin{pmatrix} \frac{H}{\sqrt{gH}} & -\frac{H}{\sqrt{gH}} \\ 1 & 1 \end{pmatrix}$$

- Now all we need is $\boldsymbol{\mathcal{V}}$

Characteristic form of the shallow water equations

- Note that (for well behaved w and $\mathcal{V}$)

$$(C_{ij})^{-1} = \left(\frac{\partial w_i}{\partial \mathcal{V}_j}\right)^{-1} = \frac{\partial \mathcal{V}_i}{\partial w_j}$$

- This means we can integrate both sides with respect to $\mathbf{w}$ to get $\boldsymbol{\nu}$

$$\boldsymbol{\nu} = \begin{pmatrix} \sqrt{gH} + \frac{1}{2}u \\ \sqrt{gH} - \frac{1}{2}u \end{pmatrix}$$

- This gives

$$\frac{\partial}{\partial t}\begin{pmatrix} \sqrt{gH} + \frac{1}{2}u \\ \sqrt{gH} - \frac{1}{2}u \end{pmatrix} + \begin{pmatrix} u + \sqrt{gH} & 0 \\ 0 & u - \sqrt{gH} \end{pmatrix} \frac{\partial}{\partial x}\begin{pmatrix} \sqrt{gH} + \frac{1}{2}u \\ \sqrt{gH} - \frac{1}{2}u \end{pmatrix} = \mathbf{0}$$

Summary

$$\frac{\partial}{\partial t}\begin{pmatrix}\phi\\ \phi u\end{pmatrix}+\frac{\partial}{\partial x}\begin{pmatrix}\phi u\\ \phi u^2+\frac{1}{2}\phi^2\end{pmatrix}=\mathbf{0}$$

$$\frac{\partial}{\partial t}\begin{pmatrix}\phi\\ \phi u\end{pmatrix}+\begin{pmatrix}0 & 1\\ -u^2+\phi & 2u\end{pmatrix}\frac{\partial}{\partial x}\begin{pmatrix}\phi\\ \phi u\end{pmatrix}=\mathbf{0}$$

$$\frac{\partial}{\partial t}\begin{pmatrix}H\\ u\end{pmatrix}+\begin{pmatrix}u & H\\ g & u\end{pmatrix}\frac{\partial}{\partial x}\begin{pmatrix}H\\ u\end{pmatrix}=\mathbf{0}$$

$$\frac{\partial}{\partial t}\begin{pmatrix}\sqrt{gH}+\frac{1}{2}u\\ \sqrt{gH}-\frac{1}{2}u\end{pmatrix}+\begin{pmatrix}u+\sqrt{gH} & 0\\ 0 & u-\sqrt{gH}\end{pmatrix}\frac{\partial}{\partial x}\begin{pmatrix}\sqrt{gH}+\frac{1}{2}u\\ \sqrt{gH}-\frac{1}{2}u\end{pmatrix}=\mathbf{0}$$

- Four different ways to express the same equation
- They are all used at different times, and are not an exhaustive list

Second-order PDEs

- Another way equations can be manipulated is that second-order PDEs can be obtained from a system of first order PDEs (or vice-versa)
- Consider

$$\frac{\partial}{\partial t}\begin{pmatrix} \bar{u} \\ \bar{v} \end{pmatrix} + \frac{\partial}{\partial x}\begin{pmatrix} \bar{v} \\ a^2\bar{u} \end{pmatrix} = 0$$

- If we take additional derivatives of $\bar{u}$ w.r.t. time and of $\bar{v}$ w.r.t. space, we get

$$\frac{\partial^2 \bar{u}}{\partial t^2} = a^2 \frac{\partial^2 \bar{u}}{\partial x^2}$$

- This is the **wave equation**, and we know that we can write it as a system of first-order PDEs (which may be easier to solve)
- However, working backwards is not always as straightforward - reintroducing $\bar{v}$ by integrating the wave equation is difficult

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Integral form of a differential equation

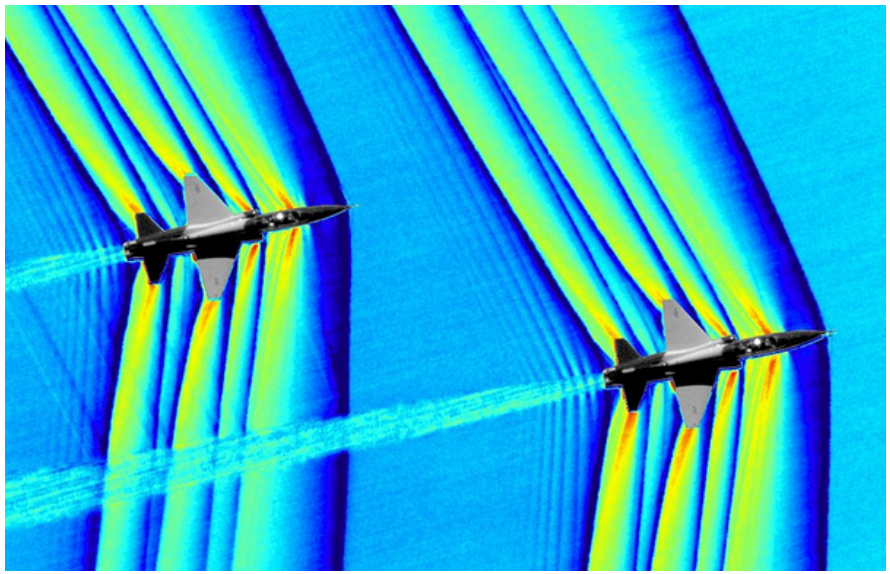
- The equations for evolving continuum descriptions of matter are typically written in differential equation form, e.g.

$$\frac{\partial u}{\partial t} + \nabla \cdot \mathbf{f}(u) = 0$$

- For this equation to be valid, we require that u is differentiable
- Mathematically, there is no reason that this is true - we shall see cases where even data which starts differentiable generates a **discontinuity**
- If this happens, we cannot solve this form of the equations, but must instead solve the **integral form**

Discontinuities

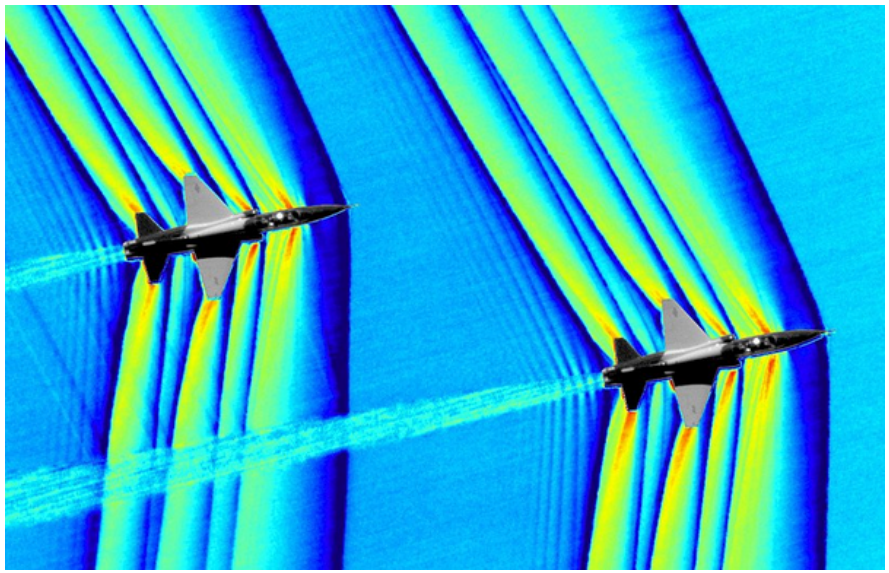
- Many discontinuities will be encountered through the continuum courses, including shock waves and material interfaces



Source: nasa.gov

- In reality, these aren't true discontinuities; there is always some form of transition

Discontinuities



Source: nasa.gov

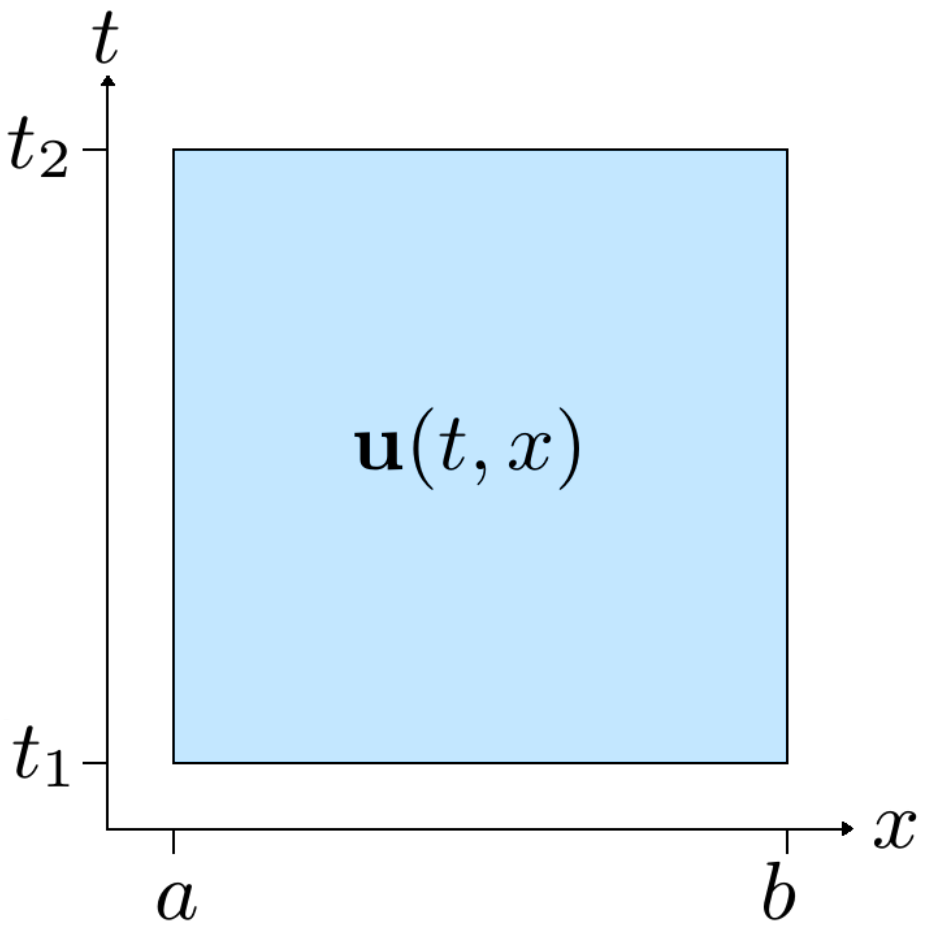
- Thermal diffusion, no matter how small, exists over shock waves
- Materials undergo a transition region (which may be atomic in scale)
- But often these are too small for a computational, or a continuum, approach to capture - equations are formulated which permit discontinuities mathematically

Integral domain

- Consider a differential equation

$$\frac{\partial u}{\partial t} + \frac{\partial f(u)}{\partial x} = 0$$

- In order to convert this to an integral equation, we first define some area we are trying to solve this equation over
- Typically, we would consider the case in which we know the solution $u(t_1, x)$, $x \in [a, b]$
- And we want to solve it for some (or all) times up until t_2, still for $x \in [a, b]$
- We can integrate our differential equation over this entire region



Integral form and conservation laws

- Begin with the assumption that u is continuous, and integrate

$$\int_a^b \int_{t_1}^{t_2} \frac{\partial u(t,x)}{\partial t} \mathrm{d}t\mathrm{d}x + \int_{t_1}^{t_2} \int_a^b \frac{\partial f(u(t,x))}{\partial x} \mathrm{d}x\mathrm{d}t = 0$$

- We can then use the fundamental theorem of calculus

$$\int_a^b u(t_2,x) - u(t_1,x)\mathrm{d}x + \int_{t_1}^{t_2} f(u(t,b)) - f(u(t,a))\mathrm{d}t = 0$$

- In this form, it would not matter if u was discontinuous, a solution could still be obtained
- For actually obtaining the solution, see "Numerical Methods for Compressible Fluid Dynamics" and "Numerical Methods for Incompressible Fluid Dynamics"

Weak solutions

- A weak solution is defined as a solution of

$$\int_a^b u(t_2, x) - u(t_1, x)\mathrm{d}x + \int_{t_1}^{t_2} f(u(t, b)) - f(u(t, a))\mathrm{d}t = 0$$

  which may, or may not, be discontinuous
- If it is discontinuous, it cannot satisfy

$$\frac{\partial u}{\partial t} + \frac{\partial f(u)}{\partial x} = 0$$

- However, this discontinuous solution is indeed a (weak) solution to the underlying physical problem, just not one that can be found by considering this form of the governing equation
- It is important to note that when solving for u on $x \in [a, b]$, $t \in [t_1, t_2]$, weak solutions **may not be unique** - we shall encounter this later